

Cloud Computing Suggestion and Assignment

1. Define cloud computing according to the NIST definition. Mention two advantages of cloud computing compared to traditional computing
2. What is a hypervisor? Give two examples.
3. State two key features of Google App Engine.
4. Briefly state the CAP theorem in context of cloud databases.
5. List any two common cloud deployment models.
6. Mention any two essential characteristics of cloud computing.
7. Define Platform as a Service (PaaS) and provide one example.
8. What is the Open Virtualization Format (OVF)?
9. List any two notable features of SQL Azure.
10. What is a Virtual Machine Monitor (VMM)?
11. Name any two types of DDoS attacks.
12. Describe load balancing in virtualized environments. Explain how Google Cloud handles load balancing.
13. Distinguish between Grid Computing, Cluster Computing, and Cloud Computing.
14. Explain the Infrastructure as a Service (IaaS) model with a real-world example.
15. Describe the layered architecture of virtualization in cloud systems.
16. Write short notes on any two Google Cloud applications, Azure Content Delivery Network, Azure AppFabric, Amazon EBS and Amazon RDS.
17. Compare Public, Private, Hybrid, and Community cloud deployment models with examples.
18. Explain the six phases of the cloud service lifecycle management process
19. Describe security challenges at the network, host, and application layers in cloud computing.
20. Explain the key characteristics of cloud computing with focus on scalability and elasticity.
21. Differentiate between IaaS, PaaS, and SaaS with suitable examples.
22. State and explain any five benefits of cloud storage.
23. Illustrate the concept of machine imaging in cloud systems.
24. Describe the lifecycle phases involved in cloud auditing outsourcing.
25. Explain how cloud systems can be integrated with wireless sensor networks and list two applications.
26. Cloud Reference Model, SOA, Virtualization, Migration and Hypervisors, seven-step cloud migration model.
27. Analyze challenges related to data security, privacy, and regulatory compliance in public cloud environments, using realistic current examples
28. cloud infrastructure architecture and its major layers.
29. Amazon EC2, Amazon S3, and Amazon SimpleDB and describe practical use-cases.
30. Auditing and compliance requirements in cloud environments and how third-party providers ensure transparency.
31. Compare: IaaS, SaaS, PaaS; Cloud vs Contemporary Computing.
32. Security concerns that may arise during virtual machine migration.